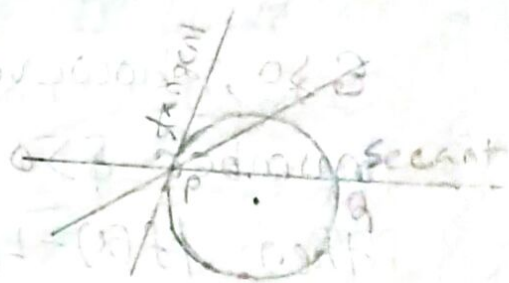


"LIMITS"

Consider two different points P and Q on the curve shown. The line through P and Q is called a secant line for the curve at P .

If we move the point Q along the curve towards P , then the secant line will rotate toward a limiting position. The line in this limiting position is called the tangent line at P .



Limit (Assertive Definition) :-

If the values of $f(x)$ can be made as close as we like to L by making x sufficiently close to a (but not equal to a), then we write,

$$\lim_{x \rightarrow a} f(x) = L$$

which is read, "The limit of $f(x)$ as x tends to a (approaches a) is L ".

Epsilon - Delta Definition:

$\lim_{x \rightarrow a} f(x) = L$, iff for any chosen number $\epsilon > 0$,

however small, there corresponds a number $\delta > 0$, such that, whenever $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon$.

Right Hand Limit:

$\lim_{x \rightarrow a^+} f(x) = L$, iff for any chosen number

$\epsilon > 0$, however small, there corresponds a number $\delta > 0$, such that, whenever $a < x < a + \delta$, then $|f(x) - L| < \epsilon$.

Left Hand Limit:

$\lim_{x \rightarrow a^-} f(x) = L$, iff for any chosen number

$\epsilon > 0$, however small, there corresponds a number $\delta > 0$, such that, whenever

$a - \delta < x < a$, then $|f(x) - L| < \epsilon$.

□ Existence of a limit at a Point :

$\lim_{x \rightarrow a} f(x)$ exist and its value will be equal to L , iff $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist and both are equal to L ,

Find $\lim_{x \rightarrow 0^-} \sqrt{x}$ and $\lim_{x \rightarrow 0^+} \sqrt{x}$ and comment if

$\lim_{x \rightarrow 0} \sqrt{x}$ exists.

Sol:

$\lim_{x \rightarrow 0^-} \sqrt{x}$ does not exist since $x < 0$ here and $\sqrt{x}$ is not defined for $x < 0$.

$\lim_{x \rightarrow 0^+} \sqrt{x}$ as we see that $\sqrt{x}$ approaches 0 as x approaches 0.

Hence, $\lim_{x \rightarrow 0} \sqrt{x}$ does not exist.

⊛ Evaluate (सीमा निकालें) —

(a) $\lim_{x \rightarrow 3^-} \frac{|x-3|}{x-3}$

(b) $\lim_{x \rightarrow 3^+} \frac{|x-3|}{x-3}$

(c) $\lim_{x \rightarrow 3} \frac{|x-3|}{x-3}$ if any (सीमा बताएं).

Sol: (a)

As x approaches 3 from the left, $x < 3$, that is, $(x-3) < 0$.

Hence, $|x-3| = -(x-3)$

Therefore, $\lim_{x \rightarrow 3^-} \frac{|x-3|}{x-3} = \frac{-(x-3)}{x-3} = -1$.

Sol: (b)

As x approaches 3 from the right, $x > 3$, that is, $(x-3) > 0$.

$$\text{Hence, } |x-3| = x-3$$

$$\text{Therefore, } \lim_{x \rightarrow 3^+} \frac{|x-3|}{x-3} = \frac{x-3}{x-3} = 1$$

Sol: (c)

As x approaches 3 from the left, $x < 3$, that is, $(x-3) < 0$.

$$\text{Hence, } |x-3| = -(x-3)$$

$$\text{Therefore, } \lim_{x \rightarrow 3^-} \frac{|x-3|}{x-3} = \frac{-(x-3)}{x-3} = -1$$

And, as x approaches 3 from the right, $x > 3$, that is, $(x-3) > 0$.

$$\text{Hence, } |x-3| = x-3$$

$$\text{Therefore, } \lim_{x \rightarrow 3^+} \frac{|x-3|}{x-3} = \frac{x-3}{x-3} = 1$$

The left hand limit and right hand limit both exist but not equal.

Hence, $\lim_{x \rightarrow 3} \frac{|x-3|}{x-3}$ does not exist.

⊛ Example □

⊛ Test the existence -

Let the limit are -

(a) $\lim_{x \rightarrow a} \frac{1}{x-a}$ (b) $\lim_{x \rightarrow a} \frac{1}{(x-a)^2}$

Sol(a)

$\lim_{x \rightarrow a^+} \frac{1}{x-a} = +\infty$ and $\lim_{x \rightarrow a^-} \frac{1}{x-a} = -\infty$

The function increases indefinitely as x approaches a from the right and decreases indefinitely as x approaches a from the left.

Thus, $\lim_{x \rightarrow a^+} \frac{1}{x-a} \neq \lim_{x \rightarrow a^-} \frac{1}{x-a}$

Hence, $\lim_{x \rightarrow a} \frac{1}{x-a}$ does not exist.

e.g.

✓ (a) $\lim_{x \rightarrow \infty} \frac{3x+1}{x}$ does this limit exist?

(b) $\lim_{x \rightarrow \infty} \frac{3x+1}{x}$, write the equation of horizontal asymptote on this limit.

Sol: a

$$\lim_{x \rightarrow \infty} \frac{3x+1}{x}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(3 + \frac{1}{x}\right)$$

$$\Rightarrow 3$$

$\therefore$ The limit is exist.

$$\checkmark \lim_{x \rightarrow +\infty} \frac{3x+1}{x}$$

$$\Rightarrow \lim_{x \rightarrow +\infty} \left(3 + \frac{1}{x}\right)$$

$$\Rightarrow 3 + 0$$

$$\Rightarrow 3$$

$$\text{and } \lim_{x \rightarrow -\infty} \frac{3x+1}{x}$$

$$\Rightarrow \lim_{x \rightarrow -\infty} \left(3 + \frac{1}{x}\right)$$

$$\Rightarrow 3 + 0$$

$$\Rightarrow 3$$

$\therefore$ Both the left hand and right hand limit
are exist and they are both equal.

Hence, the given limit is exist.

Find $\lim_{x \rightarrow -\infty} \frac{4x^2 - x}{2x^3 - 5}$

Sol Divide each term in the numerator and denominator by the highest power of x that occurs in the denominator, namely x^3 . We obtain —

$$\lim_{x \rightarrow -\infty} \frac{4x^2 - x}{2x^3 - 5} = \lim_{x \rightarrow -\infty} \frac{\frac{4}{x} - \frac{1}{x^2}}{2 - \frac{5}{x^3}}$$

$$= \frac{\lim_{x \rightarrow -\infty} \left(\frac{4}{x} - \frac{1}{x^2} \right)}{\lim_{x \rightarrow -\infty} \left(2 - \frac{5}{x^3} \right)}$$

$$= \frac{\lim_{x \rightarrow -\infty} \frac{4}{x} - \lim_{x \rightarrow -\infty} \frac{1}{x^2}}{\lim_{x \rightarrow -\infty} 2 - \lim_{x \rightarrow -\infty} \frac{5}{x^3}}$$

$$= \frac{4 \lim_{x \rightarrow -\infty} \frac{1}{x} - \lim_{x \rightarrow -\infty} \frac{1}{x^2}}{2 - 5 \lim_{x \rightarrow -\infty} \frac{1}{x^3}}$$

$$= \frac{0 - 0}{2 - 0}$$

$$= 0$$

1. Find the value of $\lim_{x \rightarrow 0} (1+x)^{1/x}$.

Sol Gr.T. $\lim_{x \rightarrow 0} (1+x)^{1/x}$

$$= \lim_{x \rightarrow 0} \left\{ 1 + \frac{1}{x} \cdot x + \frac{\frac{1}{x}(\frac{1}{x}-1)}{2} x^2 + \frac{\frac{1}{x}(\frac{1}{x}-1)(\frac{1}{x}-2)}{6} x^3 + \dots \right\}$$

$$= \lim_{x \rightarrow 0} \left\{ 1 + 1 + \frac{(1-x)}{2 \cdot x^2} \cdot x^2 + \frac{(1-x)(1-2x)}{6 \cdot x^3} \cdot x^3 + \dots \right\}$$

$$= \lim_{x \rightarrow 0} \left\{ 1 + 1 + \frac{(1-x)}{2} + \frac{(1-x)(1-2x)}{6} + \dots \right\}$$

$$= 1 + 1 + \frac{1}{2} + \frac{1}{6} + \dots$$

$$= 1 + e \quad \underline{\text{Ans}} \quad 1 + e$$

2. Evaluate $\lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x$

Gr.T., $\lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x$

Now, put $\frac{1}{x} = y$,

when $x \rightarrow \infty$, $y \rightarrow 0$

$$\therefore \lim_{y \rightarrow 0} (1+y)^{1/y}$$

$$\Rightarrow \lim_{y \rightarrow 0} \left\{ 1 + \frac{1}{y} \cdot y + \frac{\frac{1}{y}(\frac{1}{y}-1)}{2} y^2 + \frac{\frac{1}{y}(\frac{1}{y}-1)(\frac{1}{y}-2)}{6} y^3 + \dots \right\}$$

$$= \lim_{y \rightarrow 0} \left\{ 1 + 1 + \frac{1-y}{y \cdot 2} \cdot y^2 + \frac{(1-y)(1-2y)}{y^3 \cdot 6} \cdot y^3 + \dots \right\}$$

$$= \lim_{y \rightarrow 0} \left\{ 1 + 1 + \frac{1-y}{2} + \frac{(1-y)(1-2y)}{6} + \dots \right\}$$

$$= 1 + 1 + \frac{1}{2} + \frac{1}{6} + \dots$$

$$= 1 + e \quad \underline{\text{Ans}} \quad 1 + e$$

Ex. 1. Evaluate the following :

$$(i) \lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 + x} \right)$$

[C. P. 1983]

$$(ii) \lim_{x \rightarrow \infty} \left\{ x - \sqrt{(x-a)(x-b)} \right\}$$

[C. P. 1992, B. P. 1999]

Solution : (i) $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 + x} \right)$

$$= \lim_{x \rightarrow \infty} \frac{\left(x - \sqrt{x^2 + x} \right) \left(x + \sqrt{x^2 + x} \right)}{x + \sqrt{x^2 + x}}$$

$$= \lim_{x \rightarrow \infty} \frac{-x}{x \left\{ 1 + \sqrt{1 + \frac{1}{x}} \right\}}$$

$$= \lim_{x \rightarrow \infty} \frac{-1}{1 + \sqrt{1 + \frac{1}{x}}}$$

$$= -\frac{1}{2}, \quad \therefore \lim_{x \rightarrow \infty} \frac{1}{x} = 0.$$

$$(ii) \lim_{x \rightarrow \infty} \left\{ x - \sqrt{(x-a)(x-b)} \right\}$$

$$= \lim_{x \rightarrow \infty} \frac{\left\{ x - \sqrt{(x-a)(x-b)} \right\} \left\{ x + \sqrt{(x-a)(x-b)} \right\}}{x + \sqrt{(x-a)(x-b)}}$$

$$= \lim_{x \rightarrow \infty} \frac{x^2 - \{x^2 - (a+b)x + ab\}}{\left\{ x + \sqrt{(x-a)(x-b)} \right\}}$$

$$= \lim_{x \rightarrow \infty} \frac{x \left\{ -\frac{ab}{x} + (a+b) \right\}}{x \left\{ 1 + \left(1 - \frac{a}{x} \right) \left(1 - \frac{b}{x} \right) \right\}}$$

$$= \lim_{x \rightarrow \infty} \frac{(a+b) - \frac{ab}{x}}{1 + \left(1 - \frac{a}{x} \right) \left(1 - \frac{b}{x} \right)}$$

$$= \frac{1}{2}(a+b), \because \lim_{x \rightarrow \infty} \frac{ab}{x} = 0, \lim_{x \rightarrow \infty} \frac{a}{x} = 0, \lim_{x \rightarrow \infty} \frac{b}{x} = 0.$$